

Assignment #03: Animation Project – Principles and Physics

Due Date: 25th March 2026

Coursework %: approximately 45%

The purpose of this assignment is for you to showcase your understanding of the principles of animation, and to research and implement 2 physically-based animation topics.



Overview

This is an **individual** assignment (no group work).

The objective of this assignment is to demonstrate advanced understanding of the Principles of Animation alongside the investigation and application of physically based animation techniques within Unreal Engine 5. Students will create a coherent and engaging animated short story that integrates performance, storytelling, and physically grounded motion.

The assignment emphasises research-informed practice, experimentation, and critical reflection, with primary focus on technical implementation and understanding of physically based systems. Students are expected to explore how theoretical concepts from the literature relate to observed behaviour in real-time simulation, and to demonstrate thoughtful technical decision-making throughout the development process.

Description:

You are required to produce a 60–90 second animated story in Unreal Engine 5 that demonstrates clear application of animation principles through believable movement, timing, and performance. The animation should communicate a coherent narrative rather than a sequence of unrelated actions.

The animation may feature characters, objects, environments, or a combination of these elements. Both simple and complex scenarios are acceptable, provided there is clear intent and effective execution.

Tools & Assets:

Unreal Engine 5 (any version) must be used. Students may import external assets such as character models, props, or environments. However, the animation, simulation setup, and logic must be created by the student. Fully assembled scenes or projects with pre-existing animation logic should not be used.

Tutorials may be used to support learning, but submitted work must not directly replicate tutorial content. All external assets, tools, libraries, and learning resources must be clearly credited in the final report. Any assets or resources identified in the project that are not **appropriately referenced** may result in a mark of zero being awarded for the relevant component.

Submission Guidelines

1. **Deadline:** Submit your project by **25th March 2026**
 - Your submission must include:
 - A **YouTube link** to a 60-90 **second video** showing your animated story in action. The video should demonstrate the features with clear voice-over or captions.
 - A **PDF report** (example at end) containing:
 - Screenshots of your project, and description of how the principles of animation were implemented.
 - A 4-6 page concise description of **two physically based animation topics** that you researched and implemented. For each feature, you must investigate research papers or books and demonstrate an understanding of the core concepts and physics equations behind the technique.
 - Cite all resources used in your project, including research papers, books, assets, animations, and plugins. Failure to properly cite these sources will result in **0%** for the assignment.
2. **Demo Session:** You will be assigned a demo time with the demonstrator. During this time, you will show the project and explain the implementation. You must show an understanding of the Principles of Animation as well as your 2 chosen physically-based animation topics, or you will receive **0%** for the assignment.
3. **Late Submissions:** Any late submissions will result in **0%** and will be recorded as **absent** for the assignment.
4. **Original Work:** Ensure that the project is **created by you**. Using someone else's code will be considered cheating.

Requirements

Your application should be coded in Unreal Engine. Use only **free assets** from the Unreal Engine Marketplace or other free asset libraries (e.g., Mixamo, OpenGameArt). No purchased assets are allowed. You must use **Unreal Engine's animation tools**, such as Sequencer, Character Animation Blueprints, and keyframe animation. To enhance the animation, you must use 2 physically-based animation features in UE to enhance the animation, as long as it aligns with the story and principles.

The project should include the following:

1. Showcase the principles of Animation (~40%)

Your project must showcase the following **4 principles of animation** clearly. The animation should tell a coherent story. It could be a simple narrative (e.g., a character completing an action, overcoming an obstacle, or reacting to a situation), but it must have a clear beginning, middle, and end.

- **Squash and Stretch**
 - Exaggerate the shape of characters or objects to convey weight, volume, and flexibility. For example, a character squashes upon impact or stretches during take-off.
- **Anticipation**
 - Include preparatory actions before the main action (e.g., a character winding up before a jump or an object pulling back before being thrown).
- **Follow Through**
 - Secondary elements (hair, clothing, etc.) should continue moving after the main action has finished, showing natural momentum.
- **Exaggeration**
 - Push actions beyond realism to heighten emotional impact or entertainment value, e.g., exaggerated reactions or extreme squashing/stretching.

2. Physically-Based Animation Research and Implementation (~60%)

- Students must research and implement two physically-based animation techniques in Unreal Engine. This includes studying the underlying physical concepts and simulation approaches using academic or technical sources, and exploring how these ideas relate to the behaviour observed in Unreal.
- You are encouraged to experiment with engine parameters and explain how changes affect motion, linking your observations to concepts such as forces, constraints, damping, or stability. The focus is on demonstrating understanding through investigation rather than describing features.
- You do not need to analyse Unreal's internal implementation, only the general methods and relevant research.

Possible techniques to use (select 2):

- Cloth Simulation (Chaos Cloth, moderate)
- Hair / Flexible Rods (Groom + Groom Physics, complex)
- Ragdoll Physics (Chaos Physics Asset, moderate)
- Particle Systems (Niagara, simple)
- Object / Rigid Body Physics (Chaos Physics, simple)
- Smoke and Fire (Niagara Fluids, complex)
- Liquids / Fluids (Water System + Niagara Fluids, complex)
- Physics-Based Character Animation (Animation Blueprint + Physics Constraints, moderate)
- Destruction / Fracture (Chaos Destruction, complex)
- Wind / Environmental Effects (Wind Directional Source + Chaos, simple)
- Soft Body Dynamics (Chaos Flesh or third-party soft body plugin, complex)
- Buoyancy / Floating Objects (Water System Buoyancy Component, moderate)
- Gravity / Force-Based Animation (Chaos Physics + Field System, simple)
- Character-Driven Physics Interaction (Character Movement + Chaos, moderate)

Difficulty reflects the level of experimentation and interpretation required, not expected marks — strong work can be achieved with any technique.

Marks are awarded for the quality of research, depth of experimentation, and clarity of explanation in the report, as well as how effectively the techniques are implemented and integrated to support

animation principles, storytelling, and character performance. Techniques should be used intentionally and meaningfully; disconnected or purely technical effects will receive limited credit. High marks are associated with thoughtful investigation, clear reasoning, and polished, stable results.

CS7GV5 Report Example

Name:	
Project Title:	
Student ID:	
Youtube link:	

Required feature: 4 Principles of Animation
<i>Screenshot(s) of feature:</i>
<i>Describe your implementation:</i>

Physically based Animation Topic 1
<i>Screenshot(s) of feature:</i>
<i>Background Research:</i> - Describe key principles, equations, and methods from literature. - Discuss approaches and relevant developments.
<i>Implementation and Investigation in UE:</i> - Describe your setup and integration. - Show parameter exploration and experiments, including screenshots. - Discuss decisions, challenges, and observations.
<i>Evaluation and Reflection:</i> - What worked well. - Unexpected behaviours. - Limitations or trade-offs. - How physics contributed to animation principles. - What you would improve with more time.

Physically based Animation Topic 2

Screenshot(s) of feature:

Background Research:

- Describe key principles, equations, and methods from literature.
- Discuss approaches and relevant developments.

Implementation and Investigation in UE:

- Describe your setup and integration.
- Show parameter exploration and experiments, including screenshots.
- Discuss decisions, challenges, and observations.

Evaluation and Reflection:

- What worked well.
- Unexpected behaviours.
- Limitations or trade-offs.
- How physics contributed to animation principles.
- What you would improve with more time.

References: Cite research papers, books, tutorials used, links to all assets used, and Unreal Engine documentation